MAINCRAFTS

Artificial Intelligence & Machine Learning Internship

TASK 1
California Housing Price Prediction

1. Introduction

This project builds and evaluates a Linear Regression model for predicting median house values using the California Housing dataset. The workflow covers data loading, exploration, exploratory data analysis (EDA), train-test splitting, model training, prediction, evaluation, visualization, and interpretation.

2. Objective

The objective is to develop a basic house price prediction model and evaluate it using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score.

3. Dataset Description

The California Housing dataset was loaded with scikit-learn's **fetch_california_housing**. The dataset contains **20,640 observations** and **9 numerical columns**.

Feature	Role / Description
MedInc	Median income
HouseAge	Median house age
AveRooms	Average number of rooms
AveBedrms	Average number of bedrooms
Population	Population of the area
AveOccup	Average occupancy
Latitude	Geographic latitude
Longitude	Geographic longitude
MedHouseVal	Target: median house value

4. Initial Data Inspection

The dataset was inspected using DataFrame information, descriptive statistics, and a missing-value check. All nine columns contained 20,640 non-null observations. No missing values were found, so missing-value imputation was not required.

5. Exploratory Data Analysis

EDA was performed before model training. The notebook includes a target-value histogram, distributions for all numerical features, and a correlation heatmap. These visualizations help understand the data and relationships between variables.

Distribution Analysis

The feature-distribution plots show the range and shape of the numerical variables. The target distribution for **MedHouseVal** was also visualized to examine the spread of median house values.

Correlation Analysis

A correlation heatmap was generated from the numerical columns. It provides a visual summary of pairwise linear relationships between the input features and the target variable.

Missing-Value Analysis

The missing-value check returned zero missing values for every feature and for the target column.

6. Data Preparation

The target column **MedHouseVal** was separated from the eight input features. The dataset was split using an **80:20 training/testing ratio** with **random_state = 42**.

Dataset portion	Rows	Input features
Training set	16,512	8
Testing set	4,128	8

7. Linear Regression Model

A Linear Regression model from scikit-learn was trained on the training data. Predictions were then generated for the unseen testing data. The model was evaluated using MAE, RMSE, and R^2.

8. Model Evaluation Results

MAE	RMSE	R^2 Score
0.5332	**0.7456**	**0.5758**

Interpretation: The R^2 score of 0.5758 means that approximately 57.58% of the variation in the target values is explained by the fitted linear model on this test split.

9. Results and Interpretation

The model achieved a MAE of **0.5332** and an RMSE of **0.7456** on the test set. The Actual vs Predicted scatter plot was used to visually compare observed and predicted house values. The diagonal reference line represents perfect agreement between actual and predicted values.

Actual vs Predicted Analysis

Points closer to the diagonal reference line indicate smaller prediction errors, while larger deviations indicate larger errors. The plot shows a positive relationship between actual and predicted values, while also showing dispersion around the reference line. This indicates that a simple linear model does not capture all variation in the housing data.

10. Conclusion

This project demonstrates a complete basic machine learning workflow for house price prediction. The California Housing dataset was explored, checked for missing values, visualized, divided into training and testing sets, and used to train a Linear Regression model. Predictions were evaluated using MAE, RMSE, and R^2 and visualized using an Actual vs Predicted plot.

11. Possible Improvements

- Try non-linear regression models such as Random Forest Regression or Gradient Boosting.
- Perform feature engineering to create additional informative variables.
- Use cross-validation for more robust model evaluation.
- Investigate and handle influential outliers where appropriate.
- Compare multiple regression algorithms using the same evaluation metrics.
- Tune model parameters and evaluate whether more flexible models improve predictive performance.

12. Deliverables

Jupyter Notebook: task1_ml_linear_regression.ipynb
PDF Report: MainCrafts_AIML_Task1_Report.pdf

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